

Final Project Review

NeuroInsight: Resting-State fMRI Network
Classification

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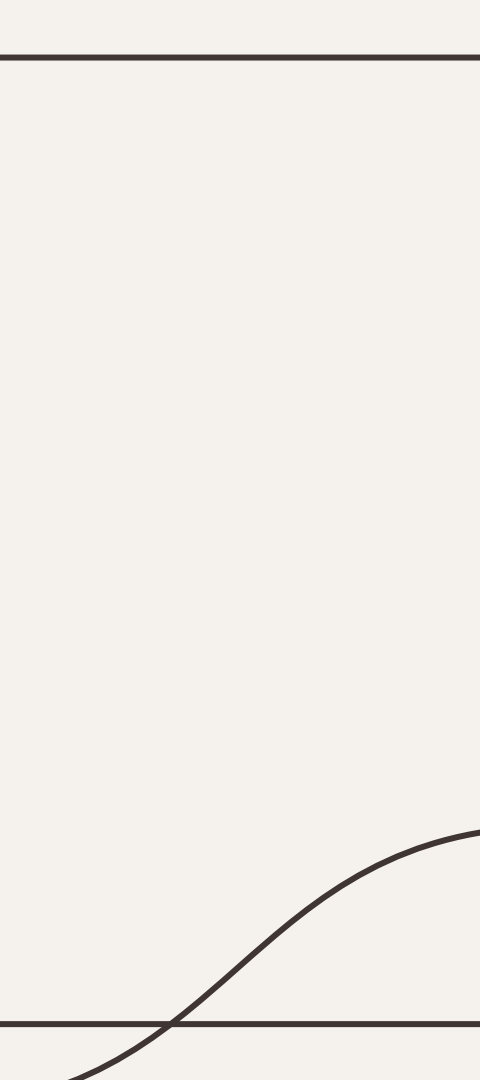


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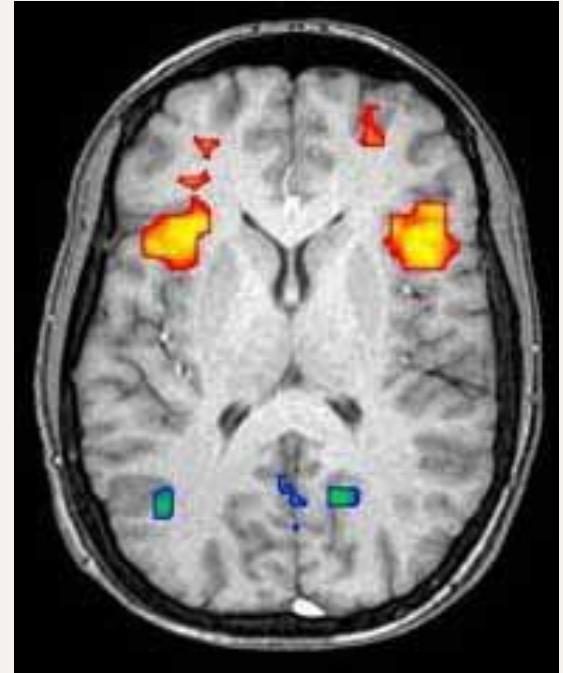
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Problem Statement and Project Goal

- Mental Health Conditions affect 1 in 4 people, but are largely invisible at the biological level.
- This project aims to develop a prototype pipeline for researchers to use to identify activation patterns in the brain and link them to aspects of mental wellbeing
- It also serves as a way for the layman to understand some of the workings of the brain

The goal of this project is to develop a machine learning model capable of analyzing stress by identifying different biomarkers and activation patterns in the brain. The project will investigate data preprocessing techniques, machine learning algorithms, and evaluation methods.



Task:

Build an end-to-end system that:

- Takes a raw resting-state fMRI scan as input
- Automatically preprocesses it into a standardised feature representation
- Predicts brain age and resting-state network activity using a trained neural network
- Produces an interpretable output a researcher can act on — not just a number, but a visual explanation of which brain regions drove the prediction
- Packages everything into an accessible interface requiring no neuroimaging expertise to operate

Action:

The project was broken down into 3 steps:

- Data and Preprocessing
- Model
- Interface and Pipeline

Each layer was built on top of or in conjunction with the previous

Data and Preprocessing

Dataset:

- MPI-LEMON dataset: 291 healthy participants
- Contains resting-state fMRI data, and is available publicly through OpenNeuro's repository
- Two features per voxel- BOLD (Blood Oxygen Level Dependent) and fALFF (fractional amplitude of low-frequency fluctuations, 0.01-0.08 Hz)

Preprocessing:

- Takes the mean BOLD and fALFF signals over the length of the scan
- The data from the dataset is resampled to a 48x48x32 picture with 2 channels: one for BOLD and one for fALFF (48,48,32,2)

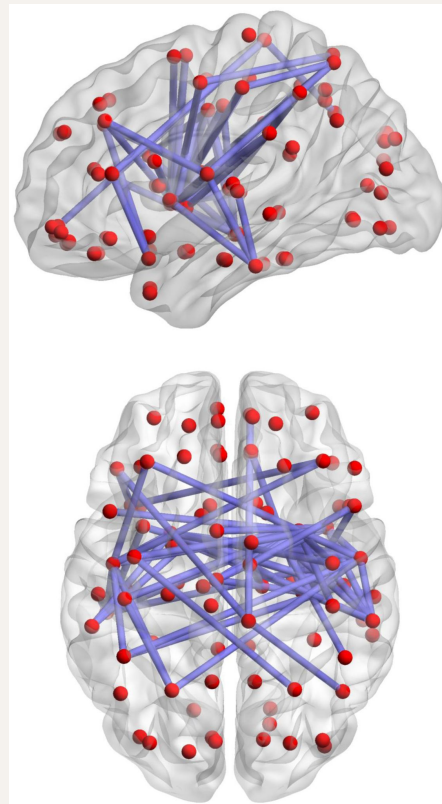
Max Planck Institut Leipzig Mind-Brain-Body Dataset - LEMON

[home](#)

Model

Overview:

- Lightweight 3D CNN (291K parameters)
 - On the smaller side to prevent overfitting
- Dual output heads: brain age regression (linear) and resting-state network classification (sigmoid)
- Three networks predicted: DMN, Salience, ECN
 - DMN: Default Mode Network
 - When you are doing nothing in particular (reflective)
 - Salience Network:
 - Deciding what the brain pays attention to (alarm system)
 - Usually hyperactive in patients reporting anxiety, elevated stress, etc.
 - ECN: Executive Control Network
 - Handles deliberate thinking (planning, making decisions)



Model Architecture

4.2 Layer Structure

Table 1: Model architecture summary.

Layer	Type	Output shape	Activation
Input	InputLayer	(48, 48, 32, 2)	—
conv3d_1	Conv3D (32 filters)	(48, 48, 32, 32)	ReLU
batch_norm_1	BatchNormalization	(48, 48, 32, 32)	—
maxpool_1	MaxPooling3D	(24, 24, 16, 32)	—
dropout_1	Dropout	(24, 24, 16, 32)	—
conv3d_2	Conv3D (64 filters)	(24, 24, 16, 64)	ReLU
batch_norm_2	BatchNormalization	(24, 24, 16, 64)	—
maxpool_2	MaxPooling3D	(12, 12, 8, 64)	—
dropout_2	Dropout	(12, 12, 8, 64)	—
conv3d_3	Conv3D (128 filters)	(12, 12, 8, 128)	ReLU
batch_norm_3	BatchNormalization	(12, 12, 8, 128)	—
global_avgpool	GlobalAveragePooling3D	(128)	—
dropout_3	Dropout	(128)	—
dense_shared	Dense (64 units)	(64)	ReLU
dropout_4	Dropout	(64)	—
<i>Wellbeing head</i>			
dense_wb	Dense (32 units)	(32)	ReLU
wellbeing_output	Dense (1 unit)	(1)	Linear
<i>Network head</i>			
dense_net	Dense (32 units)	(32)	ReLU
network_output	Dense (3 units)	(3)	Sigmoid

Front End

- React Webapp
- Uvicorn Server for Model
- Vercel for Deployment

Upload MRI Scan

Accepted formats: NIfTI (.nii / .nii.gz)



Drop files here
or click to browse your computer

Choose File

Subject details

Subject ID	Age
e.g. SUB-042	e.g. 27
Sex	Clinical Notes
Not specified	Optional

Run Analysis

Run ↗

Research use only

This tool does not constitute a clinical diagnosis. All output should be reviewed and is not meant for clinical purposes.

● Quick Test — model connected

Run Inference on a pre-processed subject from the training set (~1-3 s, no file upload needed). Use this to verify the full website pipeline before uploading a real scan.

Subject Index (0 - 290)

Run quick test →

How it works

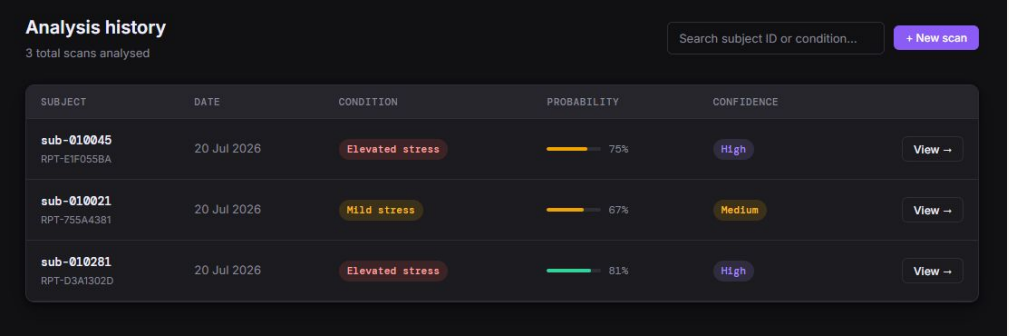
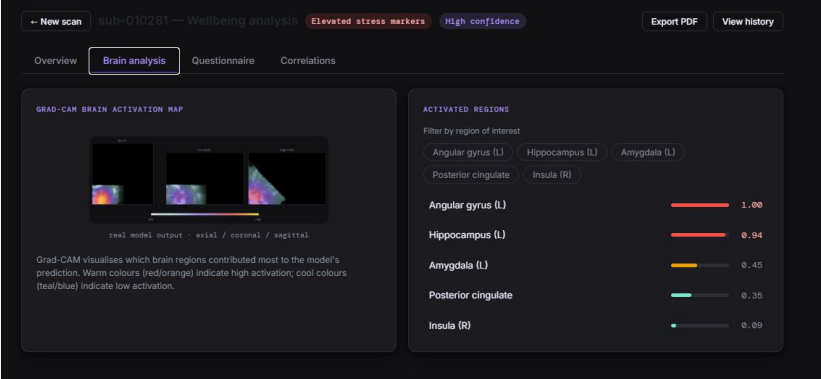
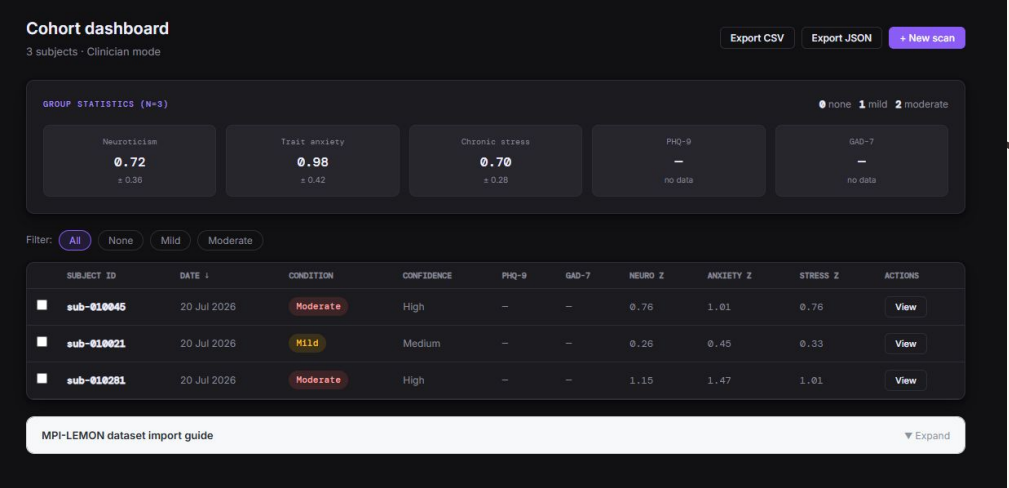
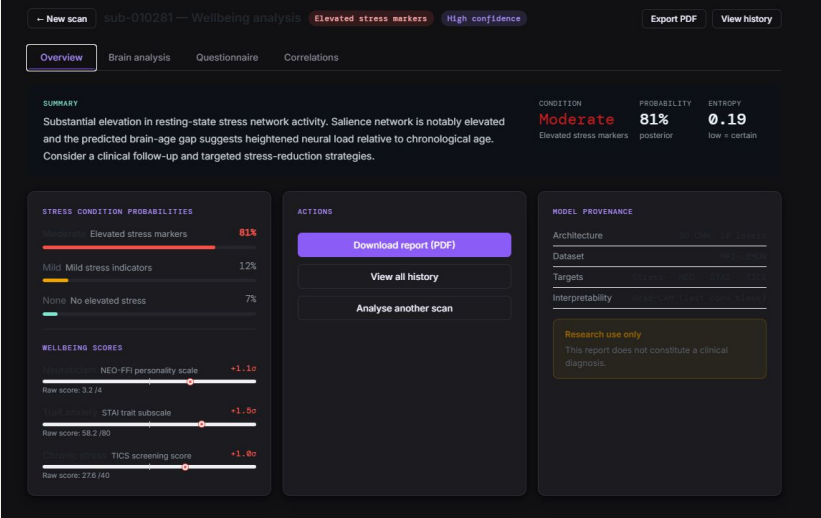
- 1 Upload your scan**

NIfTI volumes (.nii / .nii.gz). The model accepts standard resting-state fMRI output from most major scanners.
- 2 Preprocessing**

The BOLD signal is resampled, normalised, and FALFF computed for the CNN.
- 3 Prediction**

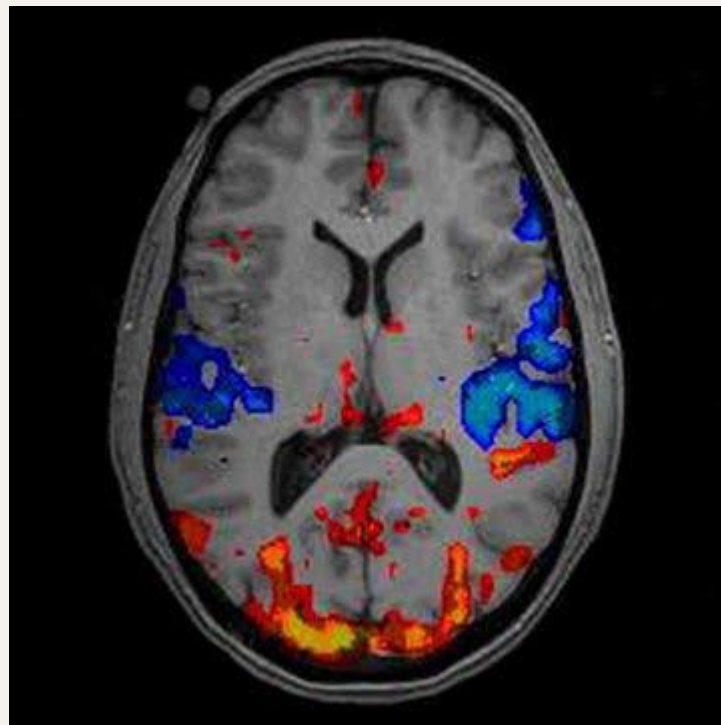
The model outputs trait anxiety, chronic stress, and neuroticism estimates with a Grad-CAM heatmap showing which regions drove the prediction.
- 4 Review and export**

Download a PDF report or save to the patient's history.



Conclusion - Limitations

- Healthy Controls Only: The MPI-LEMON dataset contains healthy adults. The model therefore has not been trained or validated on clinical populations. Stress scores are inferred from brain-age deviation and network activation, not from labelled diagnoses
- Heuristic Stress Score: The stress-indicator formula and thresholds were hand crafter, not learned. Future work should collect labelled stress data and learn the mapping end-to-end
- Small Dataset: Only 291 subjects is small for 3D CNN. It limits the total size of the model without overfitting
- Preprocessing: More advanced techniques could be used



Conclusion - Results

The model successfully learns to predict brain age from two complementary rs-fMRI features (mean BOLD and fALFF). Grad-CAM maps consistently highlight the anterior cingulate cortex, dlPFC, insula, and hippocampus — regions well-established as sensitive to stress, ageing, and psychiatric conditions. This provides face validity for the learned representations.

GitHub: <https://github.com/bawejaan12-alt/SuperSafe-Nuero-Imaging>

Research Report:

<https://drive.google.com/file/d/1RQrwGNLsjeMz5pRuXBz0JuZTvtmxcB7p/view?usp=sharing>

Learnings

- Code Organization
 - Using software like GitHub to organize changes with branches etc.
 - Helpful for a history of changes, keeping prototypes and main version separate
 - Makes it easier to test new features and ideas
- Project Planning
 - Start with the overall architecture and then build individual components
 - Started building components and figured out how to make them work together afterwards
 - Timeline and Scope
 - Having a clear timeline and scope document is essential to be able to be aware of what needs to be done at any point of time
 - All parts of the project get the attention that is needed, and it leads to a more holistic, complete, and thought through result